Complexity and a Candidate-Count-Independent Approximation Guarantee for the COMPASS-FL Selection Problem

Abstract

We study the local selection problem solved by every candidate aggregator: choose n clients out of k neighbors so as to maximize the full utility $\mathcal{U}$ (bottleneck capacity, bottleneck latency, cohesion, representativeness, fairness, and deadline adherence). We show that the problem is NP-hard, so no polynomial algorithm is exact for every k unless P=NP. We then show that a simple polynomial enumeration of *center seeds* attains a fraction $\Gamma(\mathcal{C}^\star)$ of the optimum, where Γ depends only on n, the weights, ϵ, and statistics of the optimal set itself, and not on the number k of candidates. Since 1-swap local search never decreases $\mathcal{U}$, `greedy_swap` seeded with these sets inherits the guarantee. On realistic instances the guarantee holds but is loose, while the center seeds themselves are within a few percent of the optimum.

1 Setting

Instance. A candidate aggregator a chooses a set $\mathcal{C} \subseteq \mathcal{N}$, $|\mathcal{C}| = n \geq 2$, from its neighborhood $\mathcal{N}$ of size $k = |\mathcal{N}|$. Each $i \in \mathcal{N}$ has a capacity estimate f_i, a latency $L_i = L_{i,a} < L_{\max}$, a fairness score $h_i = \min(\Delta\tau_i, \Delta\tau_{\text{cap}})/\Delta\tau_{\text{cap}} \in [0,1]$, a deadline-adherence score $Q_i \in (0,1]$, and a normalized label vector $x_i = \hat{\mathbf{y}}_i \in \mathbb{R}^K_{\geq 0}$, $\|x_i\| = 1$. Similarities are $S_{ij} = x_i^\top x_j \in [0,1]$. Let $s = \sum_{j\in\mathcal{V}} x_j$ (so $r_i = x_i^\top s$ and $\sum_{j,j'} S_{jj'} = \|s\|^2$) and, for a set $\mathcal{C}$,

$$z(\mathcal{C}) = \sum_{i\in\mathcal{C}} x_i, \qquad \mu_{\mathcal{C}} = z(\mathcal{C})/n, \qquad P(\mathcal{C}) = \sum_{i<j\in\mathcal{C}} S_{ij}.$$

Utility. With $\text{clip}(x) = \min(1, \max(\epsilon, x))$, $0 < \epsilon < 1$, the raw components are

$$\varphi_{\text{comp}} = \tfrac{\min_{\mathcal{C}} f_i}{f_{\text{ref}}}, \qquad \varphi_{\text{comm}} = 1 - \tfrac{\max_{\mathcal{C}} L_i}{L_{\max}}, \qquad \varphi_{\text{coh}} = \tfrac{P(\mathcal{C})}{\binom{n}{2}},$$
$$\varphi_{\text{repr}} = \gamma\, \tfrac{\sum_{\mathcal{C}} r_i}{\sqrt{(n+2P(\mathcal{C}))\,\|s\|^2}}, \qquad \varphi_{\text{fair}} = \tfrac{1}{n}\textstyle\sum_{\mathcal{C}} h_i, \qquad \varphi_{\text{adh}} = \tfrac{1}{n}\textstyle\sum_{\mathcal{C}} Q_i,$$

where $\gamma > 0$ is the common factor introduced by estimating $\|s\|^2$ (Eq. (repr) of the paper; $\gamma = 1$ when the estimate is exact). The clipped components are $\Phi_m = \text{clip}(\varphi_m)$ for $m \in \{\text{comp}, \text{comm}, \text{coh}, \text{repr}, \text{fair}, \text{adh}\}$, $\Phi_{\text{data}} = \text{clip}(\Phi_{\text{coh}}^{w_c}\Phi_{\text{repr}}^{w_r})$, and

$$\mathcal{U}(\mathcal{C}) = \prod_{m\in\{\text{comp,comm,data,fair,adh}\}} \Phi_m^{w_m}, \qquad w_m > 0,\ \textstyle\sum_m w_m = 1, \qquad w_c, w_r \geq 0.$$

The problem is $\max_{\mathcal{C}\subseteq\mathcal{N},|\mathcal{C}|=n} \mathcal{U}(\mathcal{C})$; $\mathcal{C}^\star$ denotes an optimal set and starred quantities refer to it ($\varphi^\star_{\text{coh}} = \varphi_{\text{coh}}(\mathcal{C}^\star)$, etc.).

Lemma 1 (aggregate form)**.** *For every $\mathcal{C}$ with $|\mathcal{C}| = n$: $\|z(\mathcal{C})\|^2 = n + 2P(\mathcal{C})$, hence*

$$\varphi_{coh}(\mathcal{C}) = \frac{\|z(\mathcal{C})\|^2 - n}{n(n-1)}, \qquad \varphi_{repr}(\mathcal{C}) = \gamma\cos\big(z(\mathcal{C}), s\big), \qquad \|\mu_{\mathcal{C}}\|^2 = \frac{1 + (n-1)\varphi_{coh}(\mathcal{C})}{n}.$$

Proof. $\|z\|^2 = \sum_{i,j\in\mathcal{C}} x_i^\top x_j = n + 2P$ since $\|x_i\| = 1$. $\sum_{\mathcal{C}} r_i = z^\top s$, so $\varphi_{\text{repr}} = \gamma z^\top s/(\|z\|\|s\|)$. The last identity divides $\|z\|^2 = n + n(n-1)\varphi_{\text{coh}}$ by n^2. □

Thus cohesion and representativeness depend on the selection only through the K-dimensional vector $z(\mathcal{C})$, the two averages through two sums, and the two bottleneck terms through two order statistics.

Lemma 2 (clipping)**.** *For $a \geq 0$ and $b > 0$:* $\text{clip}(a)/\text{clip}(b) \geq \min\{1, a/b\}$ *and* $\text{clip}(a)/\text{clip}(b) \geq \epsilon/\text{clip}(b)$.

Proof. If $a \geq b$, monotonicity of clip gives a ratio ≥ 1. If $0 < a < b$: $x \mapsto \text{clip}(x)/x$ equals ϵ/x, 1, $1/x$ on $(0, \epsilon)$, $[\epsilon, 1]$, $(1, \infty)$, so it is non-increasing and $\text{clip}(a)/a \geq \text{clip}(b)/b$. If $a = 0$ the first bound is 0. The second bound holds because $\text{clip}(a) \geq \epsilon$. □

2 Hardness

Proposition 1. *For any fixed weights with $w_c \geq w_r/2$ (in particular for $w_c + w_r = 1$, $w_c \geq 1/3$, which includes the default $w_c = w_r = 1/2$) and any $\epsilon < 1/2$, deciding whether some $\mathcal{C}$ with $|\mathcal{C}| = n$ has $\mathcal{U}(\mathcal{C}) \geq \mathcal{U}_0$ is NP-hard.*

Proof. Reduction from Clique on d-regular graphs, which remains NP-complete.[1] Given a d-regular graph $G = (V, E)$ with $d \geq 2$ and an integer $n \geq 3$, use one label per edge plus one shared label 0, and give node i the label counts $y_i[e] = 1$ for every edge e incident to i and $y_i[0] = t$, with an integer $t \geq \sqrt{d}$. Then $\|y_i\|^2 = d + t^2$ and

$$S_{ij} = \frac{A_{ij} + t^2}{d + t^2} \qquad (i \neq j),$$

where A is the adjacency matrix. Set every $f_i = f_{\text{ref}}$, $L_i = L_{\max}/2$, $\Delta\tau_i = \Delta\tau_{\text{cap}}$, $Q_i = 1$, $\gamma = 1$, and $\mathcal{N} = V$. For $|\mathcal{C}| = n$ let $e(\mathcal{C})$ be the number of edges of G inside $\mathcal{C}$ and $A(\mathcal{C}) = e(\mathcal{C}) + t^2\binom{n}{2}$. Then $P = A/(d + t^2)$ and, by Lemma 1, $\|z\|^2 = n + 2A/(d + t^2)$. Every node has $x_i^\top s = (2d + |V|t^2)/(d + t^2)$, so $z^\top s$ is the same for all $\mathcal{C}$. Hence

$$\log \mathcal{U}(\mathcal{C}) = \text{const} + w_{\text{data}}\Big[w_c \log A - \tfrac{w_r}{2} \log\big(n(d + t^2) + 2A\big)\Big],$$

provided no component is clipped. Its derivative with respect to A is

$$\frac{w_c}{A} - \frac{w_r}{n(d + t^2) + 2A} > \frac{w_c}{A} - \frac{w_r}{2A} \geq 0,$$

so $\mathcal{U}$ is strictly increasing in $e(\mathcal{C})$. No component is clipped: $\varphi_{\text{coh}} \in [t^2/(d + t^2), (1 + t^2)/(d + t^2)] \subset [1/2, 1)$; $\varphi_{\text{repr}} \geq \sqrt{t^2/(d + t^2)} \geq 1/\sqrt{2}$ (using $\|z\| \leq n$); $\Phi_{\text{data}} \geq 1/2$; $\Phi_{\text{comm}} = 1/2$; and the other components equal 1. Therefore G has a clique of size n if and only if $\max_{\mathcal{C}} \mathcal{U}(\mathcal{C}) \geq \mathcal{U}_0$, where $\mathcal{U}_0$ is the utility of a set with $e = \binom{n}{2}$. The instance has polynomial size. □

Remark 1. The construction uses $K = |E| + 1$ labels. Hardness for a *fixed* number of labels (e.g., $K = 10$) is not implied, and the problem may well be polynomial (but with a degree growing in K) in that regime.

3 Center seeds and the approximation guarantee

Definition 1 (center seed)**.** For an eligible set $E \subseteq \mathcal{N}$ and $c \in E$, the *center seed* $\mathcal{C}_c(E)$ consists of c and the $n - 1$ nodes of $E \setminus \{c\}$ with the largest $S_{c,j}$ (ties broken arbitrarily). Algorithm Center enumerates every threshold tuple $(\phi, \lambda, \alpha, \beta)$ with $\phi \in \{f_i\}$, $\lambda \in \{L_i\}$, $\alpha \in \{h_i\}$, $\beta \in \{Q_i\}$, forms $E = \{i \in \mathcal{N} : f_i \geq \phi, L_i \leq \lambda, h_i \geq \alpha, Q_i \geq \beta\}$, and, if $|E| \geq n$, evaluates $\mathcal{C}_c(E)$ for every $c \in E$. It returns the evaluated set with the largest $\mathcal{U}$.

Lemma 3 (the member nearest to the centroid)**.** *Let $\mu^\star = \mu_{\mathcal{C}^\star}$, $q^\star = \|\mu^\star\|^2 = (1 + (n-1)\varphi^\star_{coh})/n$, and let $c \in \mathcal{C}^\star$ minimize $\|x_c - \mu^\star\|$ (equivalently, maximize $\sum_{j\in\mathcal{C}^\star} S_{cj}$). Then*

$$\textit{(a)}\ \ \|x_c - \mu^\star\|^2 \leq 1 - q^\star, \qquad \textit{(b)}\ \ x_c^\top \mu^\star \geq q^\star, \qquad \textit{(c)}\ \ \sigma_c := \sum_{j\in\mathcal{C}^\star\setminus\{c\}} S_{cj} \geq (n-1)\,\varphi^\star_{coh}.$$

[1] This is a standard fact; the reference to cite should be checked by the author before the proposition is used in the paper.

Proof. For unit vectors, $\frac{1}{n}\sum_{i\in\mathcal{C}^\star}\|x_i-\mu^\star\|^2 = 1-\|\mu^\star\|^2$; the minimum is at most the average, which gives (a). Expanding (a), $1-2x_c^\top\mu^\star+q^\star \le 1-q^\star$, which is (b). Finally $\sigma_c = n\,x_c^\top\mu^\star - 1 \ge nq^\star - 1 = (n-1)\varphi^\star_{\text{coh}}$. □

Fix from now on the thresholds of the optimum, $\phi^\star = \min_{\mathcal{C}^\star} f_i$, $\lambda^\star = \max_{\mathcal{C}^\star} L_i$, $\alpha^\star = \min_{\mathcal{C}^\star} h_i$, $\beta^\star = \min_{\mathcal{C}^\star} Q_i$, the corresponding eligible set $E^\star \supseteq \mathcal{C}^\star$, the center c of Lemma 3, and $\mathcal{C}_c = \mathcal{C}_c(E^\star)$. Let $T = \sum_{j\in\mathcal{C}_c\setminus\{c\}} S_{cj}$.

Lemma 4 (cohesion)**.** *$T \ge (n-1)\varphi^\star_{coh}$ and $\varphi_{coh}(\mathcal{C}_c) \ge \max\{2\varphi^\star_{coh}-1,\ \frac{2}{n}\varphi^\star_{coh}\}$.*

Proof. $\mathcal{C}^\star\setminus\{c\}$ is an $(n-1)$-subset of $E^\star\setminus\{c\}$, and $\mathcal{C}_c\setminus\{c\}$ maximizes the sum of similarities to c over such subsets, so $T \ge \sigma_c \ge (n-1)\varphi^\star_{\text{coh}}$ by Lemma 3(c). Since $S \ge 0$, $P(\mathcal{C}_c) \ge T$, which gives $\varphi_{\text{coh}}(\mathcal{C}_c) \ge T/\binom{n}{2} \ge \frac{2}{n}\varphi^\star_{\text{coh}}$. For the other bound, let $D(\mathcal{C}) = \sum_{i<j\in\mathcal{C}}\|x_i-x_j\|^2 = n(n-1)-2P(\mathcal{C}) = n(n-1)(1-\varphi_{\text{coh}}(\mathcal{C}))$. The identity $D(\mathcal{C}) = n\sum_{i\in\mathcal{C}}\|x_i-\mu_{\mathcal{C}}\|^2$ and the fact that the centroid minimizes the sum of squared distances give

$$D(\mathcal{C}_c) \le n\sum_{i\in\mathcal{C}_c}\|x_i-x_c\|^2 = 2n(n-1-T) \le 2n(n-1)(1-\varphi^\star_{\text{coh}}),$$

that is, $1-\varphi_{\text{coh}}(\mathcal{C}_c) \le 2(1-\varphi^\star_{\text{coh}})$. □

Lemma 5 (representativeness)**.** *Let $\theta^\star = \angle(\mu^\star, s) \in [0,\pi/2]$, $\rho^\star = (1+\sqrt{2})\sqrt{1-q^\star}$ and $\beta^\star_\angle = \arcsin(\rho^\star/\sqrt{q^\star})$ if $\rho^\star < \sqrt{q^\star}$, else $\pi/2$. Then*

$$\varphi_{repr}(\mathcal{C}_c) \ge \gamma\cos\big(\min\{\pi/2,\ \theta^\star+\beta^\star_\angle\}\big).$$

Proof. By Jensen's inequality and Lemma 4, $\|\mu_{\mathcal{C}_c}-x_c\|^2 \le \frac{1}{n}\sum_{i\in\mathcal{C}_c}\|x_i-x_c\|^2 = \frac{2(n-1-T)}{n} \le \frac{2(n-1)(1-\varphi^\star_{\text{coh}})}{n} = 2(1-q^\star)$. With Lemma 3(a) and the triangle inequality, $\|\mu_{\mathcal{C}_c}-\mu^\star\| \le \rho^\star$. If $\|u-v\| \le \rho < \|v\|$, then $\angle(u,v) \le \arcsin(\rho/\|v\|)$; here $\|\mu^\star\| = \sqrt{q^\star}$. The angle is a metric on the sphere, so $\angle(\mu_{\mathcal{C}_c}, s) \le \theta^\star + \beta^\star_\angle$. All vectors are non-negative, so every angle is at most $\pi/2$, and cos is decreasing on $[0,\pi/2]$. Lemma 1 completes the proof. □

Theorem 1 (approximation, independent of k)**.** *Let*

$$\Gamma_{coh} = \max\Big\{2-\tfrac{1}{\varphi^\star_{coh}},\ \tfrac{2}{n},\ \tfrac{\epsilon}{\Phi^\star_{coh}}\Big\}\ \ (1 \text{ if } \varphi^\star_{coh}=0), \qquad \Gamma_{repr} = \max\Big\{\tfrac{\cos(\min\{\pi/2,\theta^\star+\beta^\star_\angle\})}{\cos\theta^\star},\ \tfrac{\epsilon}{\Phi^\star_{repr}}\Big\},$$

$$\kappa_{fair} = \frac{\min_{i\in\mathcal{C}^\star} h_i}{\frac{1}{n}\sum_{i\in\mathcal{C}^\star} h_i}, \qquad \kappa_{adh} = \frac{\min_{i\in\mathcal{C}^\star} Q_i}{\frac{1}{n}\sum_{i\in\mathcal{C}^\star} Q_i},$$

($\kappa_{fair} = 1$ if all $h_i = 0$ in $\mathcal{C}^\star$). Then Center *returns a set $\mathcal{C}$ with*

$$\mathcal{U}(\mathcal{C}) \ \ge\ \Gamma(\mathcal{C}^\star)\,\mathcal{U}(\mathcal{C}^\star), \qquad \Gamma(\mathcal{C}^\star) = \big(\Gamma_{coh}^{w_c}\Gamma_{repr}^{w_r}\big)^{w_{data}}\kappa_{fair}^{w_{fair}}\kappa_{adh}^{w_{adh}}.$$

Γ depends only on n, ϵ, the weights, and statistics of $\mathcal{C}^\star$; it does not depend on k or on the candidates outside $\mathcal{C}^\star$. Center *runs in time $O(k^5(k+n^2))$.*

Proof. $\mathcal{C}_c$ is among the sets evaluated by Center: the thresholds of $\mathcal{C}^\star$ are values of the enumeration, $|E^\star| \ge n$, and $c \in \mathcal{C}^\star \subseteq E^\star$. It therefore suffices to show $\mathcal{U}(\mathcal{C}_c) \ge \Gamma\,\mathcal{U}(\mathcal{C}^\star)$, and since $\mathcal{U}$ is a weighted geometric mean it suffices to bound each ratio $\Phi_m(\mathcal{C}_c)/\Phi^\star_m$.

Bottlenecks. $\mathcal{C}_c \subseteq E^\star$, so $\min_{\mathcal{C}_c} f_i \ge \phi^\star$ and $\max_{\mathcal{C}_c} L_i \le \lambda^\star$; both raw components are at least those of $\mathcal{C}^\star$ and, by monotonicity of clip, so are the clipped ones.

Averages. Every member of $\mathcal{C}_c$ has $h_i \ge \alpha^\star$ and $Q_i \ge \beta^\star$, so $\varphi_{\text{fair}}(\mathcal{C}_c) \ge \alpha^\star = \kappa_{\text{fair}}\varphi^\star_{\text{fair}}$ and likewise for adherence. Lemma 2 transfers the ratios to the clipped components ($\kappa \le 1$).

Data. By Lemmas 4, 5 and 2, $\Phi_{\text{coh}}(\mathcal{C}_c) \ge \Gamma_{\text{coh}}\Phi^\star_{\text{coh}}$ and $\Phi_{\text{repr}}(\mathcal{C}_c) \ge \Gamma_{\text{repr}}\Phi^\star_{\text{repr}}$ (for representativeness, γ cancels in the ratio of the raw values). Hence the argument of the outer clip of Φ_{data} is at least $\Gamma_{\text{coh}}^{w_c}\Gamma_{\text{repr}}^{w_r}$ times that of $\mathcal{C}^\star$, and Lemma 2 applies once more, all factors being at most 1.

Multiplying the ratios with exponents w_m gives the claim. For the running time, there are at most k^4 threshold tuples and k centers each; a seed is formed in $O(k)$ (selection) and evaluated in $O(n^2)$. □

Corollary 1 (additive utility)**.** *For $\mathcal{U}_{add} = \sum_m w_m \Phi_m$,* CENTER *returns $\mathcal{C}$ with*

$$\mathcal{U}_{add}(\mathcal{C}) \geq \min\{\Gamma_{coh}^{w_c} \Gamma_{repr}^{w_r},\, \kappa_{fair},\, \kappa_{adh}\}\, \mathcal{U}_{add}(\mathcal{C}^\star),$$

where all quantities refer to the additive optimum $\mathcal{C}^\star$.

Proof. The proof of Theorem 1 bounds each $\Phi_m(\mathcal{C}_c) \geq \Gamma_m \Phi_m^\star$ using only properties of $\mathcal{C}^\star$; sum with weights w_m. □

Corollary 2 (`greedy_swap` with center seeds)**.** *If the seeds of* ***`greedy_swap`*** *include the center seeds of* CENTER *(in addition to, or instead of, the R_s best pairs), its output $\mathcal{C}$ satisfies $\mathcal{U}(\mathcal{C}) \geq \Gamma(\mathcal{C}^\star)\,\mathcal{U}(\mathcal{C}^\star)$ for every k, because greedy completion is not needed for a seed of size n and each accepted 1-swap strictly increases $\mathcal{U}$.*

4 Remarks

Remark 2 (what the guarantee does and does not say)*.* The theorem answers how the worst-case quality depends on the number of candidates: it does not degrade as k grows, because the guarantee is expressed only in terms of the optimal set. It is, however, a worst-case bound and is loose. Γ_{repr} is non-trivial only if $\rho^\star < \sqrt{q^\star}$, i.e. $q^\star > (1+\sqrt{2})^2/(1+(1+\sqrt{2})^2) \approx 0.854$, which for $n = 10$ requires $\varphi^\star_{\text{coh}} \gtrsim 0.84$; otherwise only the clipping floor $\epsilon/\Phi^\star_{\text{repr}}$ remains. Likewise $2 - 1/\varphi^\star_{\text{coh}}$ is positive only for $\varphi^\star_{\text{coh}} > 1/2$, and κ_{fair} is small whenever the optimum mixes recently selected and long-waiting nodes.

Remark 3 (practical variant)*.* Enumerating only the capacity and latency thresholds (CENTER-2, $O(k^3(k+n^2))$) keeps the bottleneck and data bounds; the fairness and adherence factors then become $\min_{E^\star} h_i/\bar{h}^\star$ and $\min_{E^\star} Q_i/\bar{Q}^\star$, which depend on the candidate pool.

Remark 4 (noisy similarities)*.* With CKKS and differential-privacy noise, the matrix $\tilde{S}$ used by the protocol is not exactly a Gram matrix, which the proofs of Lemmas 3–5 use. If $|\tilde{S}_{ij} - S_{ij}| \leq \eta$, the selection step of $\mathcal{C}_c$ loses at most $2(n-1)\eta$ in T, so Lemma 4 holds with an additional -4η; a full perturbation analysis of the theorem is left open.

5 Numerical check

The script `approximation_bound.py` (in `experiments/`) evaluates the theorem on the 200 instances of `selection_optimality.py` ($k = 50$, $n = 10$, $N = 1000$, Dirichlet $\alpha = 0.3$, default weights), whose optima were certified by branch and bound. For every instance it computes $\Gamma(\mathcal{C}^\star)$, the ratio $\mathcal{U}(\mathcal{C}_c)/\mathcal{U}^\star$ of the seed used in the proof, and the ratios of CENTER-2 without and with 1-swap local search.

quantity	min	median	mean	max
$\Gamma(\mathcal{C}^\star)$ (Theorem 1)	0.243	0.394	0.391	0.436
$\mathcal{U}(\mathcal{C}_c)/\mathcal{U}^\star$ (seed of the proof)	0.95	1	0.996	1
CENTER-2 / $\mathcal{U}^\star$	0.973	0.994	0.992	1
CENTER-2 + 1-swap / $\mathcal{U}^\star$	0.984	1	0.999	1
$\varphi^\star_{\text{coh}}$	0.266	0.395	0.393	0.556
Γ_{coh}	0.2	0.2	0.2	0.201
Γ_{repr}	0.00103	0.00112	0.00113	0.0014
κ_{fair}	0.0676	0.741	0.735	1
κ_{adh}	0.567	0.853	0.864	1

The inequality $\mathcal{U}(\mathcal{C}_c) \geq \Gamma\,\mathcal{U}^\star$ held in all 200 instances. On these instances the optimum has cohesion between 0.27 and 0.56, so $\Gamma_{\text{coh}} = 2/n$ and Γ_{repr} is reduced to its clipping floor $\epsilon/\Phi^\star_{\text{repr}}$. The resulting Γ (median 0.39) is driven mostly by the small elasticities $w_{\text{data}} w_c = w_{\text{data}} w_r = 0.1$: the trivial bound obtained by replacing the whole data factor by its clipping floor, $(\epsilon/\Phi^\star_{\text{data}})^{w_{\text{data}}} \kappa_{\text{fair}}^{w_{\text{fair}}} \kappa_{\text{adh}}^{w_{\text{adh}}}$, already gives a median of 0.25. In contrast, the seed of the proof reaches at least 0.950 of the optimum (median 1), CENTER-2 at least 0.973, and CENTER-2 followed by 1-swap at least 0.984. The guarantee should therefore be presented as a qualitative statement (the worst case does not degrade with k), not as an estimate of the gap, which is measured directly; the center seeds themselves are a strong and cheap initialization.